Speed and velocity – what's the difference? 9a

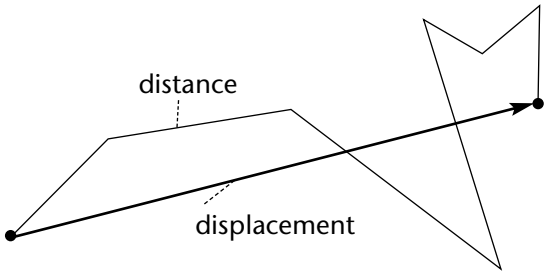
1 Read through the following paragraphs. Cross out the incorrect terms in brackets.

Speed and velocity (mean / means) almost the same (thing / think) but (their / there) is a slight distinction between the (too / two) terms. For example, if we write that a car (travelled / travels) at a constant (rate / rates) of 60 (km/h / kph), we are specifying (its / it's) speed. But if we (say / says) that the car (are travelling / is travelling) at 60 (kmh / km/h) to the (south / South), we (are talking / is talking) about velocity. Therefore, when we (specify / spicify) the speed (and / or) the direction, we (are / is) talking about (speed / velocity). Both speed (and / or) velocity (are / is) represented by v.

speed: v = 60 km/h velocity: v = 60 km/h south

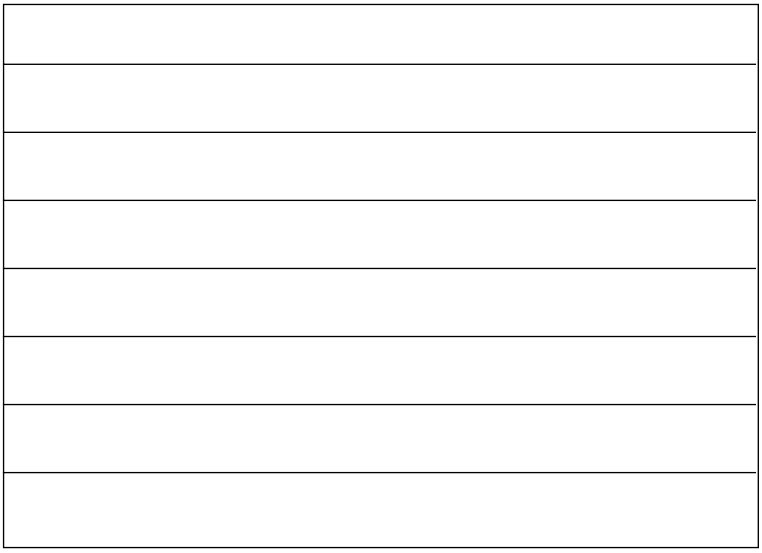
A similar distinction (exist / exists) between distance and displacement. Distance (are / is) about how much ground (has been covered / has covered) during a trip, while displacement (staits /states) how far out of place an object (is / was). In other words, distance (has / is) the travelled length of the trip and displacement (are / is) about the object's change in position. Displacement (tells / tels) how far the object (are / is) away from its original position and in which direction the new place (is / was). Displacement (is / was) the shortest distance between start (but / and) finish. The symbol for distance (has / is) d and the symbol for displacement is s.

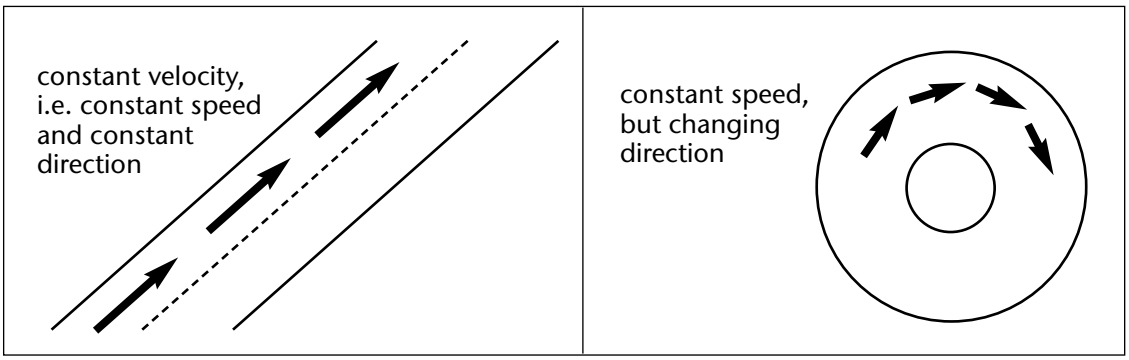
Consequently, speed (calculates / is calculated) by dividing the distance the object has travelled by the time taken to travel this distance. Velocity (finds / is found) by dividing the object's displacement by the time taken to make the displacement.



2 Cut out the sentence fragments on page 25, sort them into their correct order and then glue them into the spaces below. There are two paragraphs, a and b.

Paragraph a





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Paragraph b

3 Draw lines to join each concept to its correct symbol, unit and formula.

Concept	Symbol	Unit	Formula
displacement	d	m	d/t
distance	s	m west	s/t
speed	t	s	s/v
time	v	m/s	vt
velocity	v	m/s east	vt

4 Define each of the following:

a distance _____

b displacement _____

c speed _____

d velocity _____

e average speed _____

f instantaneous speed _____

5 Draw ticks or crosses in the cells to show whether the features of size and direction are associated with each concept.

Concept	Has size?	Has direction?
distance		
displacement		
speed		
average speed		
instantaneous speed		
velocity		
average velocity		
instantaneous velocity		

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Speed and velocity – what's the difference? 9c

6 Complete the following table by relating distance, displacement, time, speed and velocity where appropriate.

Distance/ Displacement	Time	Speed	Velocity
60 km	2 h		
1 500 m E	30 s		
120 km S		40 km/h	
400 m		80 m/s	
	5 h		50 km/h N
	40 s		20 m/s W

Sentence fragments to be cut out for Question 2

Paragraph a:

along a curved path like a roundabout. But

Constant speed and constant velocity also are

for example, a car driving along a long straight

its direction is changing continually in the roundabout.

moving at constant speed in the same direction;

not the same. Constant velocity describes an object

path. However, a car can move at a constant speed

this car does not have constant velocity, because

Paragraph b

by the time it took to cover that distance. Instantaneous speed

downhill, your instantaneous speed might well have been 30 km/h.

example, imagine you were cycling to a friend's house. It took you

is the speed at which an object is travelling at a particular time. For

km/h. But when you stopped at the

lights, your instantaneous speed was 0 km/h, and when you raced

one hour to travel 10 kilometres, so your average speed is 10

Sometimes average speed and instantaneous speed are confused. Average

speed is found by dividing the travelled distance

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